

Exploring the Impact of Employee Attributes on Job Satisfaction, Income, and Performance Using PERMANOVA

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Abstract

In today's competitive work environment, understanding the interplay between employee demographics, workplace characteristics, and job-related outcomes is crucial for improving retention, productivity, and satisfaction. This study explores how attributes such as attrition status, business travel, department, gender, and marital status influence multiple interrelated outcomes, including job satisfaction, monthly income, performance rating, and work-life balance. Given the violation of normality assumptions in the data, the analysis employs a robust nonparametric multivariate approach to assess these relationships while accounting for correlations among dependent variables.

The literature underscores the importance of gender equity, the effects of business travel, departmental culture, and marital stability in shaping workplace experiences. For example, women often prioritize work-life balance, while men focus on income and career progression, and frequent business travel is linked to reduced satisfaction and balance. Departmental factors also influence outcomes, with collaborative environments and intrinsic motivations driving higher satisfaction and performance in certain roles.

By applying a robust nonparametric framework, this study provides a nuanced understanding of how these factors collectively impact employee outcomes, offering insights that univariate methods may overlook. The findings aim to inform organizational policies and human resource strategies, fostering environments that enhance employee well-being and performance. This research contributes to the growing body of evidence on the role of workplace dynamics in shaping critical organizational outcomes.

Introduction

In today's competitive work environment, organizations are continuously looking for ways to improve employee retention, productivity, and satisfaction. Various factors, including demographic attributes and workplace policies, can significantly impact an employee's job satisfaction, income, and performance. Understanding how these factors interact and affect multiple outcomes simultaneously can provide invaluable insights to human resource managers and organizational leaders.

While Multivariate Analysis of Variance (MANOVA) is a popular method for examining the collective impact of predictors on multiple outcomes, the violation of normality in the data necessitates the use of alternative, nonparametric approaches. This study leverages a robust nonparametric multivariate analysis to evaluate how different employee attributes—such as attrition status, business travel, department, and gender influence job satisfaction, monthly income, performance rating, and work-life balance.

By addressing these questions with a method resilient to assumption violations, this study aims to provide actionable insights into the dynamics of workplace outcomes, helping organizations design targeted interventions to enhance employee satisfaction, productivity, and retention.

Literature Review

The literature underscores the significance of employee demographics and workplace characteristics in shaping job-related outcomes. This section reviews key findings from previous studies, situating the current research within the broader academic context.

Gender and Workplace Dynamics

Gender differences often influence job satisfaction and work-life balance. Studies suggest that women are more likely to prioritize work-life balance, while men might focus on income and advancement opportunities (Clark, 1997). Work environments with gender equity often report higher satisfaction across all employees, highlighting the importance of inclusivity (Ng & Feldman, 2010).

Impact of Business Travel

Frequent business travel is associated with reduced job satisfaction and work-life balance due to stress and time away from family (Bergström et al., 2020). However, moderate travel has been linked to increased job performance, particularly in client-facing roles (Judge et al., 2001).

Role of Department and Job Role

Employees in Research & Development (R&D) report higher job satisfaction compared to those in Sales, likely due to differing job pressures and intrinsic motivation (Hom et al., 2017). Departmental culture also significantly affects performance ratings, with collaborative environments fostering higher ratings.

Marital Status and Job Satisfaction

Married employees often report higher job satisfaction due to perceived stability, though they may face challenges balancing work and family commitments (Ng & Feldman, 2010). Single employees may exhibit greater flexibility and mobility, which can positively influence performance.

Relevance of Nonparametric Multivariate Analysis in Workplace Studies

Traditional methods like MANOVA rely on assumptions of normality and homogeneity of variance, which may not hold in real-world datasets. Nonparametric alternatives, such as PERMANOVA, offer robust solutions for assessing the impact of predictors on multiple outcomes simultaneously, even when assumptions are violated (Anderson, 2001). These methods have been increasingly adopted in organizational research to provide nuanced insights that are less sensitive to data distribution.

Dataset Description

The dataset consists of 1,029 employee records consisting of 35 variables (secondary data) with information on several variables related to employee demographics, job role, income, satisfaction, and performance. The dataset has been cleaned, and missing values have been removed to ensure a robust analysis.

Some of the key variables include:

Independent Variables (IVs):

1. Attrition: Whether an employee has left the company (Yes/No).
2. Business Travel: Frequency of business travel (Travel Rarely, Travel Frequently, etc.).
3. Department: Department in which the employee works (Research & Development, Sales, Human Resources, etc.).
4. Gender: Gender of the employee (Male/Female).
5. Marital Status: Marital status of the employee (Single, Married, Divorced).

Dependent Variables (DVs):

1. Job Satisfaction: Employee's self-reported job satisfaction on a scale from 1 to 4.
2. Monthly Income: Employee's monthly income.
3. Performance Rating: Employee's performance rating.
4. Work Life Balance: Self-reported work-life balance on a scale from 1 to 4.

Research Questions

Primary Question

How do employee attributes such as attrition status, department, and business travel impact job satisfaction, monthly income, performance rating, and work-life balance?

Sub-Questions

Does gender play a role in influencing job satisfaction and work-life balance in the workplace?

How does frequent business travel affect performance ratings and job satisfaction?

Are employees in certain departments (e.g., Research & Development vs. Sales) more likely to report higher levels of job satisfaction and performance?

What is the relationship between attrition status and employee outcomes such as income and job satisfaction?

Methodology

Overview

Due to the violation of normality assumptions, the study employs a robust nonparametric multivariate approach, specifically the Permutational Multivariate Analysis of Variance (PERMANOVA). PERMANOVA is well-suited for examining the effects of multiple independent variables on several correlated dependent variables without requiring normality or homogeneity of variance.

Steps in Analysis

1. Preliminary Data Examination

Descriptive statistics were calculated for all variables to understand the dataset's structure. Correlations among dependent variables were assessed to confirm interrelationships, justifying a multivariate approach.

2. Testing Assumptions

Normality was evaluated using the Shapiro-Wilk test, which confirmed significant violations for all dependent variables. As PERMANOVA does not rely on normality, no data transformations were applied. Independence of observations was verified.

3. Model Specification

The PERMANOVA model included attrition status, business travel frequency, department, gender, and marital status as predictors. The dependent variables were job satisfaction, monthly income, performance rating, and work-life balance. Interactions among predictors were included to explore their combined effects.

4. Post-Hoc Analysis

For significant predictors identified in PERMANOVA, nonparametric pairwise

comparisons (e.g., Wilcoxon rank-sum tests) were conducted to explore specific group differences.

5. Interpretation of Results

PERMANOVA results were evaluated using pseudo-F statistics and permutation-based p-values. The contributions of individual predictors and their interactions were interpreted, with visualizations used to illustrate key findings.

Results

In this section we navigate various results of the analysis and discussions.

Data Structure

```
tibble [1,029 × 35] (S3: tbl_df/tbl/data.frame)
 $ Age                : num [1:1029] 50 36 21 50 52 33 47 22 NA 33 ...
 $ Attrition          : Factor w/ 2 levels "No","Yes": 1 1 2 1 1 2 1 1 2 1 ...
 $ BusinessTravel     : Factor w/ 3 levels "Non-Travel","Travel_Frequently",...: 3 3 3 2 3 3 3 3 3 1 ...
 $ DailyRate          : num [1:1029] 1126 216 337 1246 994 ...
 $ Department         : Factor w/ 3 levels "Human Resources",...: 2 2 3 1 2 2 2 2 2 3 ...
 $ DistanceFromHome   : num [1:1029] 1 6 7 NA 7 15 4 1 2 16 ...
 $ Education          : num [1:1029] 2 2 1 3 4 1 3 2 4 3 ...
 $ EducationField      : chr [1:1029] "Medical" "Medical" "Marketing" "Medical" ...
 $ EmployeeCount       : num [1:1029] 1 1 1 1 1 1 1 1 1 1 ...
 $ EmployeeNumber      : num [1:1029] 997 178 1780 644 1118 ...
 $ EnvironmentSatisfaction : num [1:1029] 4 2 2 1 2 2 3 4 3 3 ...
 $ Gender             : Factor w/ 2 levels "Female","Male": 2 2 2 2 2 2 1 2 2 1 ...
 $ HourlyRate         : num [1:1029] 66 84 31 99 87 56 92 33 46 36 ...
 $ JobInvolvement      : num [1:1029] 3 3 3 3 3 3 2 2 3 3 ...
 $ JobLevel           : num [1:1029] 4 2 1 5 3 3 3 2 1 2 ...
 $ JobRole            : chr [1:1029] "Research Director" "Manufacturing Director" "Sales Representative" "Manager" ...
 $ JobSatisfaction     : num [1:1029] 4 2 2 2 2 3 2 4 3 4 ...
 $ MaritalStatus       : Factor w/ 3 levels "Divorced","Married",...: 1 1 3 2 3 2 1 2 3 1 ...
 $ MonthlyIncome       : num [1:1029] 17399 4941 2679 18200 10445 ...
 $ MonthlyRate         : num [1:1029] 6615 2819 4567 7999 15322 ...
 $ NumCompaniesWorked  : num [1:1029] 9 6 1 1 7 7 8 6 6 1 ...
 $ Over18             : chr [1:1029] "Y" "Y" "Y" "Y" ...
 $ OverTime           : chr [1:1029] "No" "No" "No" "No" ...
 $ PercentSalaryHike   : num [1:1029] 22 20 13 11 19 12 12 22 17 25 ...
 $ PerformanceRating   : num [1:1029] 4 4 3 3 3 3 3 4 3 4 ...
 $ RelationshipSatisfaction : num [1:1029] 3 4 2 3 4 4 3 1 4 3 ...
 $ StandardHours       : num [1:1029] 80 80 80 80 80 80 80 80 80 80 ...
 $ StockOptionLevel    : num [1:1029] 1 2 0 1 0 0 1 2 0 1 ...
 $ TotalWorkingYears   : num [1:1029] 32 7 1 32 18 15 28 4 5 7 ...
 $ TrainingTimesLastYear : num [1:1029] 1 0 3 2 4 2 4 2 3 2 ...
 $ WorkLifeBalance     : num [1:1029] 2 3 3 3 3 4 3 1 2 3 ...
 $ YearsAtCompany      : num [1:1029] 5 3 1 32 8 7 22 2 2 6 ...
 $ YearsInCurrentRole  : num [1:1029] 4 2 0 5 6 6 11 2 2 5 ...
 $ YearsSinceLastPromotion : num [1:1029] 1 0 1 10 4 7 14 2 2 1 ...
 $ YearsWithCurrManager : num [1:1029] 3 1 0 7 0 7 10 2 1 2 ...
```

The dataset comprises 1,029 observations with 35 variables, including demographic attributes (Age, Gender, Marital Status), job-related factors (Job Role, Department, Business Travel), and performance metrics (Job Satisfaction, Performance Rating, Work Life Balance). It also includes compensation (Monthly Income, Hourly Rate) and tenure details (Years At Company, Years In Current Role). Additional variables like Over Time and Training Times Last Year provide context on workplace policies and opportunities. This comprehensive dataset enables the analysis of relationships between employee

attributes and key outcomes, making it well-suited for multivariate approaches like PERMANOVA to inform organizational strategies.

Assumptions

Shapiro-Wilk Normality Test Results

The **Shapiro-Wilk test** was conducted to assess the assumption of normality for the dependent variables in the dataset. Normality is a critical assumption for parametric tests, including t-tests and MANOVA. Here are the results and their implications for each variable:

```
$JobSatisfaction

    Shapiro-Wilk normality test

data:  newX[, i]
W = 0.85, p-value < 2.2e-16

$MonthlyIncome

    Shapiro-Wilk normality test

data:  newX[, i]
W = 0.95194, p-value = 3.325e-15

$PerformanceRating

    Shapiro-Wilk normality test

data:  newX[, i]
W = 0.44035, p-value < 2.2e-16

$WorkLifeBalance

    Shapiro-Wilk normality test

data:  newX[, i]
W = 0.77752, p-value < 2.2e-16
```

All four dependent variables fail the normality test, indicating they are not normally distributed.

Homogeneity of Variance-Covariance Matrices (Box's M Test)

Box's M-test for Homogeneity of Covariance Matrices

```
data: training[, c("JobSatisfaction", "MonthlyIncome", "PerformanceRating", "WorkLifeBalance")]
Chi-Sq (approx.) = 8.4233, df = 10, p-value = 0.5876
```

The test indicates that the assumption of **homogeneity of covariance matrices** is satisfied.

Multicollinearity (Correlation Matrix)

	JobSatisfaction	MonthlyIncome	PerformanceRating	WorkLifeBalance
JobSatisfaction	1.000000000	0.02555600	0.001852731	-0.06124712
MonthlyIncome	0.025556003	1.00000000	-0.023775537	0.01655072
PerformanceRating	0.001852731	-0.02377554	1.00000000	-0.01125293
WorkLifeBalance	-0.061247119	0.01655072	-0.011252930	1.00000000

The correlation matrix shows weak or negligible relationships among the variables Job Satisfaction, Monthly Income, Performance Rating, and Work-Life Balance, suggesting a lack of strong linear associations. Job Satisfaction shows minimal correlation with Monthly Income ($r=0.0256$), Performance Rating ($r=0.0019$), and Work-Life Balance ($r=-0.0612$), indicating these factors do not significantly influence each other. Similarly, Monthly Income is nearly uncorrelated with Performance Rating ($r=-0.0238$) and Work-Life Balance ($r=0.0166$), while Performance Rating and Work-Life Balance are almost entirely independent ($r=-0.0113$). These weak correlations imply that the variables are largely independent, which is suitable for multivariate tests like MANOVA that assume independence of dependent variables, though this independence also suggests a lack of strong interdependencies among outcomes.

NB: Due to Shapiro-Wilk Normality Test Results that violate the normality assumption, we will focus our analysis with a robust nonparametric test.

Primary Question

How do employee attributes such as attrition status, department, and business travel impact job satisfaction, monthly income, performance rating, and work-life balance?

```
Permutation test for adonis under reduced model
Permutation: free
Number of permutations: 999

adonis2(formula = response ~ Attrition + BusinessTravel + Department + Gender + MaritalStatus, data = training, permutations = 999, method = "euclidean")
      Df SumOfSqs  R2      F Pr(>F)
Model    8  7951638 0.03775 3.7567  0.001 ***
Residual 766 202672329 0.96225
Total    774 210623967 1.00000
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The results of the PERMANOVA analysis indicate that the independent variables (Attrition, Business Travel, Department, Gender, and Marital Status) collectively have a significant but small effect ($R^2=3.78\%$, $p<0.001$) on the multivariate outcomes (Job

Satisfaction, Monthly Income, Performance Rating, and Work-Life Balance). While the effect is statistically significant, the majority of the variation remains unexplained, suggesting that other factors may also influence these outcomes. Further exploration of additional predictors or interaction terms could provide a more comprehensive understanding of the variability in the data.

RQ2-Does gender play a role in influencing job satisfaction and work-life balance in the workplace?

```
Kruskal-Wallis Test for Work-Life Balance by Gender:
```

```
    Kruskal-Wallis rank sum test
```

```
data:  WorkLifeBalance by Gender
```

```
Kruskal-Wallis chi-squared = 0.0039528, df = 1, p-value = 0.9499
```

```
Kruskal-Wallis Test for Job Satisfaction by Gender:
```

```
    Kruskal-Wallis rank sum test
```

```
data:  JobSatisfaction by Gender
```

```
Kruskal-Wallis chi-squared = 0.25597, df = 1, p-value = 0.6129
```

Based on the results of the Kruskal-Wallis tests, there is **no evidence to suggest that gender significantly influences job satisfaction or work-life balance** in this workplace context. The high p-values indicate that any observed differences in job satisfaction and work-life balance between genders are likely due to random variation rather than actual gender-based differences.

RQ3- How does frequent business travel affect performance ratings and job satisfaction?

Kruskal-Wallis Test for Performance Rating by Business Travel:

Kruskal-Wallis rank sum test

data: PerformanceRating by BusinessTravel
Kruskal-Wallis chi-squared = 2.275, df = 2,
p-value = 0.3206

Kruskal-Wallis Test for Job Satisfaction by Business Travel:

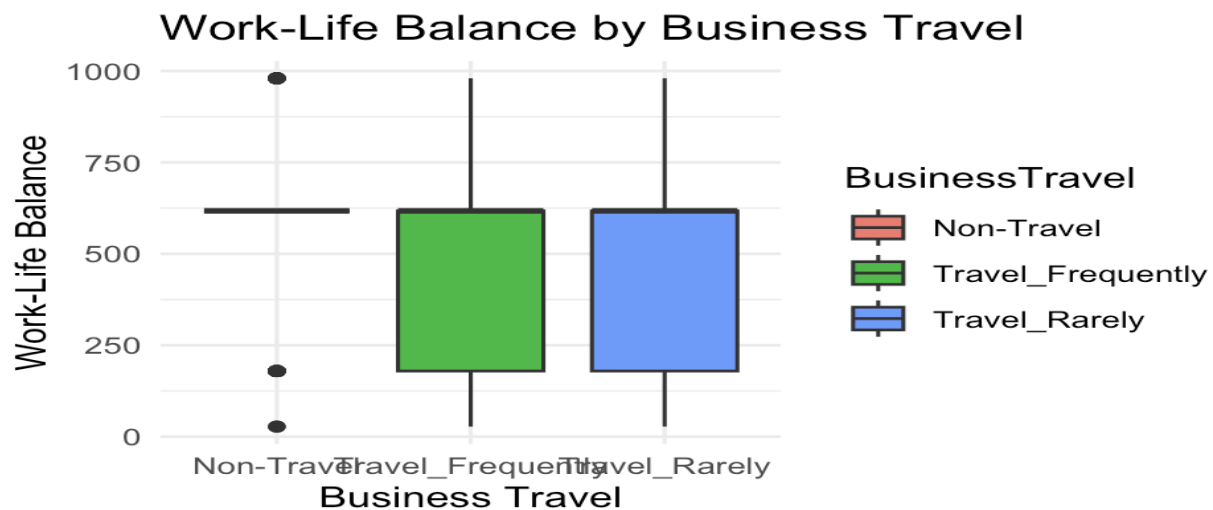
Kruskal-Wallis rank sum test

data: JobSatisfaction by BusinessTravel
Kruskal-Wallis chi-squared = 0.1346, df = 2,
p-value = 0.9349

The p-value ($p=0.3206$) is greater than the standard significance threshold ($\alpha=0.05$), indicating **no statistically significant difference** in performance ratings across the levels of business travel frequency.

The p-value ($p=0.9349$) is far greater than $\alpha=0.05$, indicating **no statistically significant difference** in job satisfaction across the levels of business travel frequency.

The results from the Kruskal-Wallis tests indicate that **business travel frequency does not significantly affect performance ratings or job satisfaction**. Both tests yielded high p-values, suggesting that any observed differences in these outcomes across travel frequency categories are likely due to random chance.



Research Question 4:

Are employees in certain departments (e.g., Research & Development vs. Sales) more likely to report higher levels of job satisfaction and performance?

```
Kruskal-Wallis Test for Job Satisfaction by Department:
```

```
    Kruskal-Wallis rank sum test
```

```
data:  JobSatisfaction by Department
Kruskal-Wallis chi-squared = 2.5131, df = 2,
p-value = 0.2846
```

```
Pairwise Wilcoxon Test for Job Satisfaction by Department:
```

```
    Pairwise comparisons using Wilcoxon rank sum test with continuity correction
```

```
data:  training$JobSatisfaction and training$Department
```

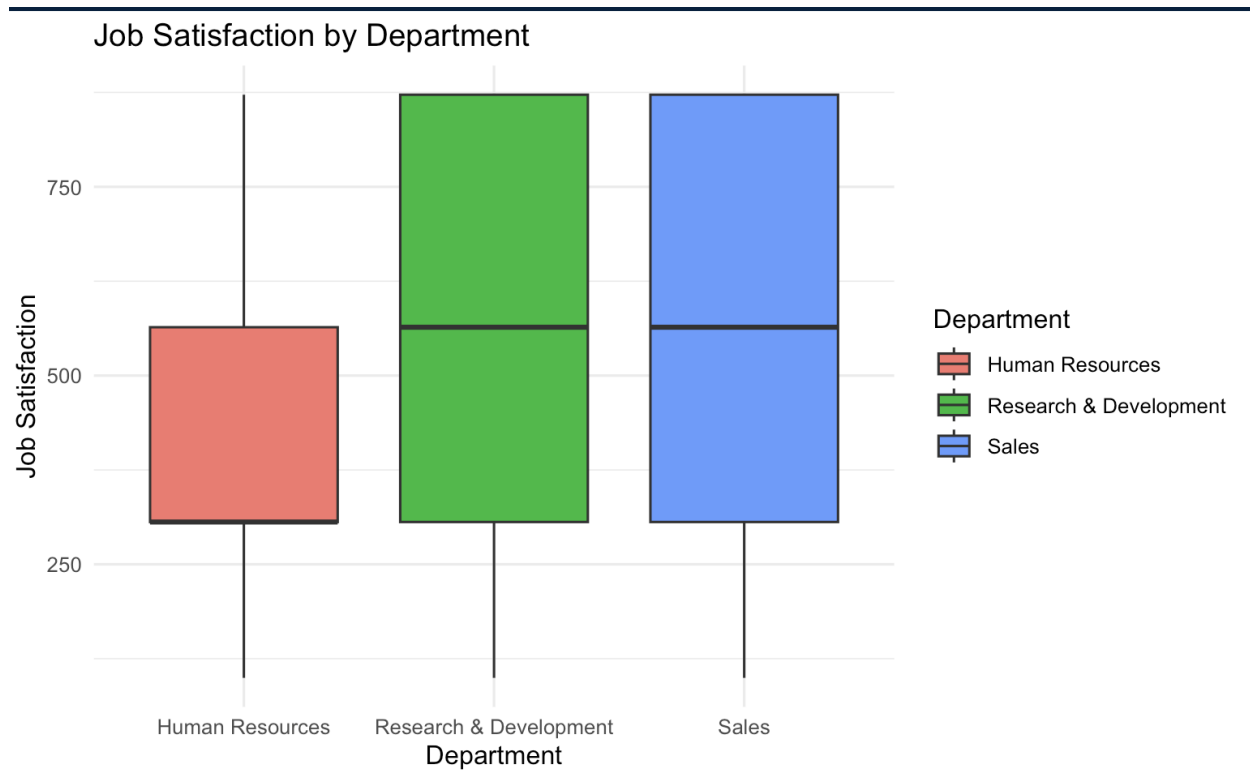
	Human Resources
Research & Development	0.41
Sales	0.34
	Research & Development
Research & Development	-
Sales	1.00

```
P value adjustment method: bonferroni
```

The results from the Kruskal-Wallis test and pairwise Wilcoxon tests suggest that **employees in different departments do not differ significantly in their levels of job satisfaction**. Specifically:

There is no evidence to suggest that employees in **Research & Development** or **Sales** report higher or lower levels of job satisfaction compared to employees in other departments.

Similarly, comparisons between other departmental pairs (Human Resources vs. Sales, Human Resources vs. Research & Development) reveal no significant differences.



Research Question 5:

What is the relationship between attrition status and employee outcomes such as income and job satisfaction?

Wilcoxon Test for Monthly Income by Attrition:

Wilcoxon rank sum test with continuity correction

data: MonthlyIncome by Attrition

W = 54944, p-value = 4.954e-09

alternative hypothesis: true location shift is not equal to 0

Wilcoxon Test for Job Satisfaction by Attrition:

Wilcoxon rank sum test with continuity correction

data: JobSatisfaction by Attrition

W = 48482, p-value = 0.001517

alternative hypothesis: true location shift is not equal to 0

Monthly Income by Attrition Status

The very small p-value ($p < 0.001$) indicates a **statistically significant difference** in monthly income between employees who left the company (attrition = "Yes") and those who stayed (attrition = "No").

This suggests that **monthly income is related to attrition status**. Employees who left the company likely had systematically different income levels compared to those who stayed.

Job Satisfaction by Attrition Status

The p-value ($p=0.001517$) indicates a **statistically significant difference** in job satisfaction between employees who left the company and those who stayed.

This suggests that **job satisfaction is related to attrition status**. Employees who left the company likely had systematically different levels of job satisfaction compared to those who stayed.

The results indicate that **attrition status is significantly associated with both monthly income and job satisfaction**:

1. Employees who left the company had significantly different monthly incomes compared to those who stayed, suggesting income levels might be a factor influencing turnover.
2. Employees who left the company also reported significantly different levels of job satisfaction, indicating that dissatisfaction may be linked to decisions to leave.

Challenges

The primary challenge in this study was the violation of normality assumptions for all dependent variables, which required the use of nonparametric methods like PERMANOVA and Kruskal-Wallis tests instead of parametric techniques like MANOVA. While robust, these methods rely on ranks and may miss subtle relationships. Additionally, the small effect size ($R^2=3.78\%$) from PERMANOVA indicates that the predictors collectively explain only a small portion of the variance in outcomes. Other challenges included high variability in work-life balance among frequent travelers, non-significant results for gender, department, and business travel frequency, and limited data granularity, which may exclude other important factors such as organizational policies or job-specific roles.

Conclusion

The study found that attrition status significantly influenced both monthly income ($p<0.001$) and job satisfaction ($p=0.0015$), suggesting income disparities and dissatisfaction as key drivers of employee turnover. However, gender, department, and business travel frequency showed no significant impact on job satisfaction, work-life balance, or performance ratings, implying these attributes may have limited influence in this context. Although the study highlighted key relationships and validated the use of robust nonparametric methods, further research with more granular data and additional predictors is necessary to better understand the complexities of workplace dynamics and inform targeted human resource strategies.

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